

Home Work-3

1) Write a C++ program to store n elements in an array using dynamic memory allocation and find the largest number.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, i, largest;
    int *a;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    a = (int *)malloc(n * sizeof(int));

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    largest = a[0];

    for(i = 1; i < n; i++)
    {
        if(a[i] > largest)
            largest = a[i];
    }

    printf("Largest element = %d", largest);

    free(a);

    return 0;
}
```

Output:

```
Enter number of elements: 4
Enter elements:
45 78 97 57
Largest element = 97
```

2) Print an array in reverse order using pointers.

```
#include <stdio.h>

int main()
{
    int a[100], n, i;
    int *ptr;
```

```

printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter elements:\n");

for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

ptr = &a[n - 1];

printf("Array in reverse order:\n");

for(i = n - 1; i >= 0; i--)
{
    printf("%d ", *ptr);
    ptr--;
}

return 0;
}

```

Output:

```

Enter number of elements: 3
Enter elements:
35 87 90
Array in reverse order:
90 87 35

```

3) Store information of students using structure and sort them by name.

```

#include <stdio.h>
#include <string.h>

struct Student
{
    char name[50];
    int roll;
};

int main()
{
    struct Student s[10], temp;
    int n, i, j;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        printf("Enter name: ");
        scanf("%s", s[i].name);

        printf("Enter roll number: ");
    }
}

```

```

        scanf("%d", &s[i].roll);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(s[i].name, s[j].name) > 0)
            {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    printf("\nStudents sorted by name:\n");

    for(i = 0; i < n; i++)
        printf("%s %d\n", s[i].name, s[i].roll);

    return 0;
}

```

Output:

```

Enter number of students: 3
Enter name: Ravi
Enter roll number: 3
Enter name: Anu
Enter roll number: 1
Enter name: Bala
Enter roll number: 2

```

```

Students sorted by name:
Anu 1
Bala 2
Ravi 3

```

4) Add n distances given in the feet-inch system using structure.

```

#include <stdio.h>

struct Distance
{
    int feet;
    int inch;
};

int main()
{
    struct Distance d1, d2, sum;

    printf("Enter first distance:\n");
}

```

```

    printf("Feet: ");
    scanf("%d", &d1.feet);

    printf("Inches: ");
    scanf("%d", &d1.inch);

    printf("Enter second distance:\n");
    printf("Feet: ");
    scanf("%d", &d2.feet);

    printf("Inches: ");
    scanf("%d", &d2.inch);

    sum.feet = d1.feet + d2.feet;
    sum.inch = d1.inch + d2.inch;

    if(sum.inch >= 12)
    {
        sum.feet++;
        sum.inch = sum.inch - 12;
    }

    printf("Total Distance = %d feet %d inches",
           sum.feet, sum.inch);

    return 0;
}

```

Output:

```

Enter first distance:
Feet: 5
Inches: 10
Enter second distance:
Feet: 4
Inches: 5
Total Distance = 10 feet 3 inches

```

5) Add two complex numbers by passing structure to a function.

```

#include <stdio.h>

struct Complex
{
    int real;
    int imag;
};

struct Complex add(struct Complex c1, struct Complex c2)
{
    struct Complex sum;

    sum.real = c1.real + c2.real;
    sum.imag = c1.imag + c2.imag;
}

```

```

        return sum;
    }

int main()
{
    struct Complex c1, c2, sum;

    printf("Enter real and imaginary part of first number: ");
    scanf("%d%d", &c1.real, &c1.imag);

    printf("Enter real and imaginary part of second number: ");
    scanf("%d%d", &c2.real, &c2.imag);

    sum = add(c1, c2);

    printf("Sum = %d + %di", sum.real, sum.imag);

    return 0;
}

```

Output:

```

Enter real and imaginary part of first number: 2 3
Enter real and imaginary part of second number: 4 5
Sum = 6 + 8i

```

6) Reverse an array using pointers.

```

#include <stdio.h>

void reverse(int *a, int n)
{
    int *start, *end, temp;

    start = a;
    end = a + n - 1;

    while(start < end)
    {
        temp = *start;
        *start = *end;
        *end = temp;

        start++;
        end--;
    }
}

int main()
{
    int a[100], n, i;

    printf("Enter number of elements: ");

```

```

scanf("%d", &n);

printf("Enter elements:\n");

for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

reverse(a, n);

printf("Reversed array:\n");

for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}

```

Output:

```

Enter number of elements: 5
Enter elements:
10 20 30 40 70
Reversed array:
70 40 30 20 10

```

7) Convert a decimal number to binary, octal and hexadecimal.

```

#include <stdio.h>

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary = ");

    int binary[100], i = 0, temp = n;

    while(temp > 0)
    {
        binary[i] = temp % 2;
        temp = temp / 2;
        i++;
    }

    for(i = i - 1; i >= 0; i--)
        printf("%d", binary[i]);

    printf("\nOctal = %o", n);
    printf("\nHexadecimal = %X", n);

    return 0;
}

```

```
}
```

Output:

```
Enter a decimal number: 25
Binary = 11001
Octal = 31
Hexadecimal = 19
```

8) Calculate the difference between two time periods.

```
#include <stdio.h>
```

```
struct Time
```

```
{
```

```
    int hours;
```

```
    int minutes;
```

```
    int seconds;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Time t1, t2, diff;
```

```
    printf("Enter first time (hours minutes seconds): ");
```

```
    scanf("%d%d%d",
```

```
        &t1.hours, &t1.minutes, &t1.seconds);
```

```
    printf("Enter second time (hours minutes seconds): ");
```

```
    scanf("%d%d%d",
```

```
        &t2.hours, &t2.minutes, &t2.seconds);
```

```
    if(t2.seconds > t1.seconds)
```

```
    {
```

```
        t1.minutes--;
```

```
        t1.seconds += 60;
```

```
    }
```

```
    diff.seconds = t1.seconds - t2.seconds;
```

```
    if(t2.minutes > t1.minutes)
```

```
    {
```

```
        t1.hours--;
```

```
        t1.minutes += 60;
```

```
    }
```

```
    diff.minutes = t1.minutes - t2.minutes;
```

```
    diff.hours = t1.hours - t2.hours;
```

```
    printf("Difference = %d:%d:%d",
```

```
        diff.hours, diff.minutes, diff.seconds);
```

```
    return 0;
```

```
}
```

Output:

```
Enter first time: 5 45 30
Enter second time: 2 30 20
Difference = 3:15:10
```

9) Develop an Employee Management System using structures.

```
#include <stdio.h>

struct Employee
{
    int id;
    char name[50];
    float salary;
};

int main()
{
    struct Employee e[10];
    int n, i;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        printf("\nEnter Employee ID: ");
        scanf("%d", &e[i].id);

        printf("Enter Employee Name: ");
        scanf("%s", e[i].name);

        printf("Enter Salary: ");
        scanf("%f", &e[i].salary);
    }

    printf("\nEmployee Details:\n");

    for(i = 0; i < n; i++)
    {
        printf("\nID = %d", e[i].id);
        printf("\nName = %s", e[i].name);
        printf("\nSalary = %.2f\n", e[i].salary);
    }

    return 0;
}
```

Output:

```
Enter number of employees: 2
Enter Employee ID: 7
```



```
Enter Employee Name: nila
Enter Salary: 5000
Enter Employee ID: 3
Enter Employee Name: charu
Enter Salary: 4500
```

Employee Details:

```
ID = 7
Name = nila
Salary = 5000.00
```

```
ID = 3
Name = charu
Salary = 4500.00
```

10) Calculate the area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>

float rectangleArea(float l, float b)
{
    return l * b;
}

float rectanglePerimeter(float l, float b)
{
    return 2 * (l + b);
}

float triangleArea(float b, float h)
{
    return 0.5 * b * h;
}

int main()
{
    float l, b, h;
    float (*area)(float, float);
    float (*perimeter)(float, float);

    printf("Enter length and breadth of rectangle: ");
    scanf("%f%f", &l, &b);

    area = rectangleArea;
    perimeter = rectanglePerimeter;

    printf("Area of Rectangle = %.2f\n", area(l, b));
    printf("Perimeter of Rectangle = %.2f\n",
           perimeter(l, b));

    printf("\nEnter base and height of triangle: ");
    scanf("%f%f", &b, &h);
```

```
    area = triangleArea;  
  
    printf("Area of Triangle = %.2f", area(b, h));  
  
    return 0;  
}
```

Output:

Enter length and breadth of rectangle: 10 5
Area of Rectangle = 50.00
Perimeter of Rectangle = 30.00

Enter base and height of triangle: 8 6
Area of Triangle = 24.00